

MATH 4570 - Matrix Methods in Data Analysis and Machine Learning

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§3. Linear Spaces over Fields

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$\{V, +, \text{scalar prod.}, X\} \Leftrightarrow \text{algebras}$

" associative algebra
" non-
" Lie algebra
" commutative algebra

1. Linear Spaces

$\mathbb{R}, \mathbb{Z}, \mathbb{C}$

mg $\mathbb{Z}$ module

$\mathbb{R}^1$

$\mathbb{R}^3$

Definition 1. Let $\mathbb{F}$ be a field. A **vector space** over $\mathbb{F}$ is any nonempty set V with two **closed** operations,

- **Sum.** $\vec{u} + \vec{v} \in V$
- **Scalar product.** $c \cdot \vec{u} \in V$

subject to **axioms**:

- 1.) $\vec{u} + \vec{v} = \vec{v} + \vec{u}$.
- 2.) $(\vec{u} + \vec{v}) + \vec{w} = \vec{v} + (\vec{u} + \vec{w})$.
- 3.) $\exists$ a zero vector $\vec{0} \in V$ s.t. $\vec{u} + \vec{0} = \vec{u}$.
- 4.) $\forall \vec{u} \in V, \exists$ a vector $-\vec{u} \in V$ s.t. $\vec{u} + (-\vec{u}) = \vec{0}$.
- 5.) $c \cdot (\vec{u} + \vec{v}) = c \cdot \vec{u} + c \cdot \vec{v}$.
- 6.) $(c + d) \cdot \vec{u} = c \cdot \vec{u} + d \cdot \vec{u}$.
- 7.) $c(d \cdot \vec{u}) = (cd)\vec{u}$.
- 8.) $1 \cdot \vec{u} = \vec{u}$.

$\{V, +\}$ is a abelian group.

Proposition 2. (1) Zero vector is unique.
(2) For any $\vec{u}$, the inverse vector $-\vec{u}$ is unique.

Proposition 3. $\vec{0} \cdot \vec{u} = \vec{0}$,

$\uparrow \quad \uparrow \quad \uparrow$
 $\mathbb{F} \quad V \quad V$

Proposition 4. $c \cdot \vec{0} = \vec{0}$

Proposition 5. $-\vec{u} = (-1)\vec{u}$.

$\mathbb{R}$

Example 6. Vectors over $\mathbb{F}$:

1. $\mathbb{F}^n$ is a vector space over $\mathbb{F}$. $\mathbb{R}^n$
2. The set of all $m \times n$ matrices (denoted as $\mathbb{F}^{m \times n}$ or $M_{m \times n}$) with matrix sum and scalar product is a vector space with entries in $\mathbb{F}$.
3. The set of all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ with sum and scalar product is a vector space.
 $e^x, \sin x + \cos x, x + \lg x, \dots$ $f+g$ $2f(x)$
4. The set of all continuous (or smooth) functions $f: \mathbb{R} \rightarrow \mathbb{R}$ with sum and scalar product is a vector space.
5. The set of all maps $\mathbb{R}^m \rightarrow \mathbb{R}^n$ is a vector space.

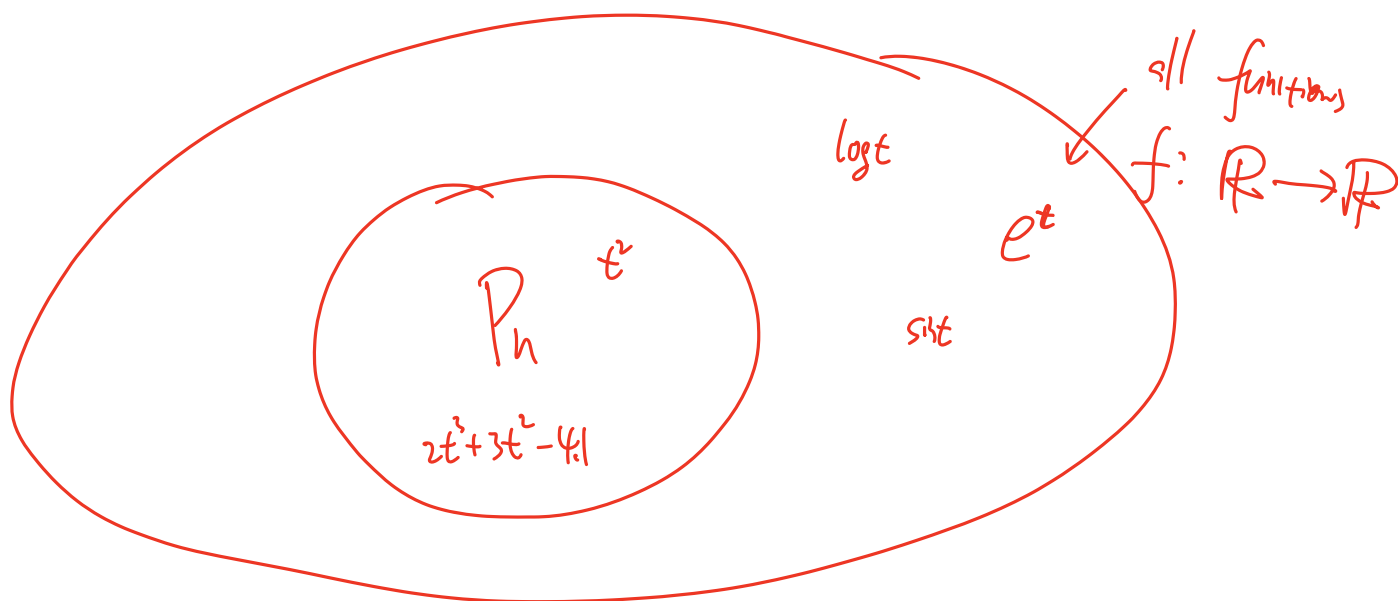
2. Subspaces

Definition 7. A subspace of the vector space V is a subset H of V that satisfies the following three properties.

- (1). $\vec{0} \in H$. $\leftarrow$ non-empty
- (2). If $\vec{u}, \vec{v} \in H$ then $\vec{u} + \vec{v} \in H$. Sum is "closed"
- (3). If $\vec{u}, \vec{v} \in H$ and $c \in \mathbb{F}$, then $c\vec{u} \in H$. scalar prod. is "closed"

Theorem 8. A subspace H of a linear space V is a linear space.

Proposition 9. $\{\vec{0}\}$ is a subspace of linear space V , called zero space.



Example 10. Determine which of the following set is a subspace (vector space).

1. Let P_n be the set of all polynomials of degree $\leq n$, (over $\mathbb{R}$). Yes

Proof. The set of polynomials of degree $\leq n$, $P_n = \{a_0 + a_1t + \cdots + a_nt^n \mid a_i \in \mathbb{R}\}$ is a subset of the linear space of all functions $F \rightarrow F$.

First, $0 \in P_n$.

Second, For any two such polynomials $a_0 + a_1t + \cdots + a_nt^n$ and $b_0 + b_1t + \cdots + b_nt^n \in P_n$, the sum $a_0 + a_1t + \cdots + a_nt^n + b_0 + b_1t + \cdots + b_nt^n = (a_0 + b_0) + (a_1 + b_1)t + \cdots + (a_n + b_n)t^n$ is in P_n .

Third, for any polynomials $a_0 + a_1t + \cdots + a_nt^n \in P_n$ and any $c \in F$, the scalar product $c(a_0 + a_1t + \cdots + a_nt^n) = ca_0 + ca_1t + \cdots + ca_nt^n$ is in P_n .

So P_n is a subspace of the linear space of all functions $F \rightarrow F$. $\square$

2. Let P be the set of all polynomials. Yes.

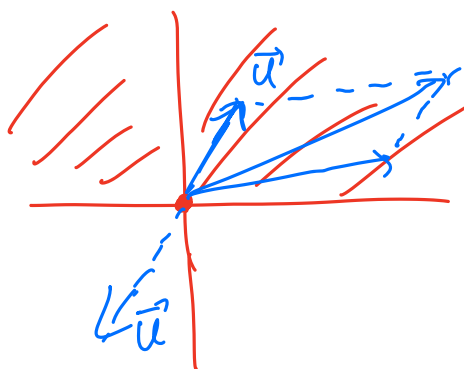
3. Let H be the set of all polynomials of degree exactly 3. NO.

4. The set $D_{n \times n}$ of all $n \times n$ diagonal matrices with real entries. $\begin{bmatrix} a_1 & & \\ & \ddots & \\ & & a_n \end{bmatrix}$

5. The set of all $n \times n$ invertible matrices with real entries. NO

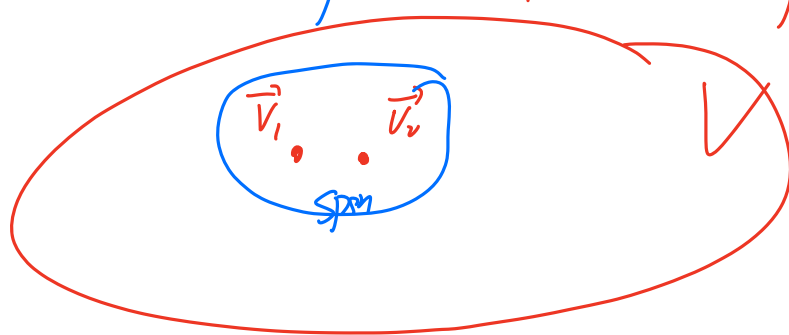
6. The union of the first and second quadrants in the xy -plane:

$$W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \mid y \geq 0 \right\} \quad \text{NO}$$



$$7. H = \{ a_1 \sin x + a_2 e^x \mid a_1, a_2 \in \mathbb{R} \} \subset \{ f: \mathbb{R} \rightarrow \mathbb{R} \}$$

$$= \text{Span} \{ \sin x, e^x \}$$



Definition 11. A **linear combination** of $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m$ in V is a vector in V defined as

$$c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_m \vec{v}_m$$

The **span** of $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m$ is the set of all linear combinations

$$\text{Span}(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m) := \left\{ c_1 \vec{v}_1 + \dots + c_m \vec{v}_m \mid c_i \in \mathbb{R} \right\}$$

Theorem 12. Then $\text{Span}(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m)$ is a subspace of V .

• Any Span is a subspace.

Proof. We prove the theorem by verifying the definition.

1. Choose all $c_i = 0$ so $\vec{0} \in \text{Span}(\vec{u}_1, \vec{u}_2, \dots, \vec{u}_m)$

2. For any two vectors $c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_m \vec{v}_m$ and $d_1 \vec{v}_1 + d_2 \vec{v}_2 + \dots + d_m \vec{v}_m$ in $\text{Span}(\vec{u}_1, \vec{u}_2, \dots, \vec{u}_m)$, the sum

$$c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_m \vec{v}_m + d_1 \vec{v}_1 + d_2 \vec{v}_2 + \dots + d_m \vec{v}_m = (c_1 + d_1) \vec{v}_1 + (c_2 + d_2) \vec{v}_2 + \dots + (c_m + d_m) \vec{v}_m$$

is an element in $\text{Span}(\vec{u}_1, \vec{u}_2, \dots, \vec{u}_m)$.

3. For any vector $c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_m \vec{v}_m$ in $\text{Span}(\vec{u}_1, \vec{u}_2, \dots, \vec{u}_m)$ and any $k \in \mathbb{F}$, the scalar product

$$k(c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_m \vec{v}_m) = kc_1 \vec{v}_1 + kc_2 \vec{v}_2 + \dots + kc_m \vec{v}_m$$

is an element in $\text{Span}(\vec{u}_1, \vec{u}_2, \dots, \vec{u}_m)$. □

• **Proposition 13.** Any subspace U of V can be written as span of some vectors in V .

If a vector space V can be written as a span of finite number of vectors in V , then V is called a **finite-dimensional** vector space.

$V = \{ \text{all functions: } \mathbb{R}' \rightarrow \mathbb{R}' \}$
infinite-dimensional.

$$\begin{array}{ccc} V \times V & \xrightarrow{+} & V \\ \downarrow T \times T & & \downarrow T \\ V \times V & \xrightarrow{+} & V \end{array}$$

3. Linear transformations

Let V and W be vector spaces over a field $\mathbb{F}$. A transformation T from V to W is a rule

$$T: V \rightarrow W$$

$$T(c_1 \vec{u} + c_2 \vec{v}) = c_1 T(\vec{u}) + c_2 T(\vec{v}).$$

Definition 14. A transformation $T: V \rightarrow W$ is called linear if

$$\textcircled{1} T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$$

$$\textcircled{2} T(c \cdot \vec{u}) = c T(\vec{u})$$

for any $c \in \mathbb{F}$
 $\vec{u}, \vec{v} \in V$

Proposition 15. If $T: V \rightarrow W$ is a linear transformation, then

$$\textcircled{1} T(\vec{0}) = \vec{0}$$

$$\textcircled{2} T(c_1 \vec{u}_1 + \dots + c_n \vec{u}_n) = c_1 T(\vec{u}_1) + \dots + c_n T(\vec{u}_n)$$

Example 16. 1. Zero map is linear transformation. $0: V \rightarrow W$
 $\vec{v} \rightarrow \vec{0}$

2. Identity map $\text{id}: V \rightarrow V$ is a linear transformation.
 $\vec{v} \rightarrow \vec{v}$

Example 17. Is $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$, defined as $T(\vec{x}) = \begin{bmatrix} x_1 + 2x_2 + 1 \\ x_2 - x_3 \end{bmatrix}$ a linear transformation?
NO! because $T(\vec{0}) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \neq \vec{0}$

Proposition 18. If $T: X \rightarrow Y$ is linear and has a inverse S , then S is linear.

Definition 19. Two vector spaces V and W are called isomorphic, denoted as

$$V \cong W$$

if there is an invertible linear transformation $T: V \rightarrow W$.

Let V and W be vector spaces. Denote $\mathcal{L}(V, W)$ the set of all linear transformations from V to W . Thus $T \in \mathcal{L}(V, W)$ if and only if

- (i) $T : V \rightarrow W$,
- (ii) $T(\vec{v}_1 + \vec{v}_2) = T(\vec{v}_1) + T(\vec{v}_2)$ for $\vec{v}_1, \vec{v}_2 \in V$,
- (iii) $T(a\vec{v}) = aT(\vec{v})$ for $\vec{v} \in V, a \in \mathbb{F}$.

$$T_2 : V \rightarrow W$$

$$T_1 : V \rightarrow W$$

Theorem 20. $\mathcal{L}(V, W)$ is a vector space.

Theorem 21. $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ is isomorphic to the matrix space $\mathbb{R}^{m \times n}$.

$$\begin{array}{ccc} \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m) & \xrightarrow{\quad} & \mathbb{R}^{m \times n} \\ T_A & \xleftrightarrow{S} & A \end{array}$$

$$T_A(\vec{x}) = A\vec{x}$$

Definition 22. The vector space $V^* := \mathcal{L}(V, \mathbb{R})$ is called the **dual space** of the vector space V .

Cohomology

• Algebraic Topology

4. Kernel and Image

Definition 23. Consider a transformation

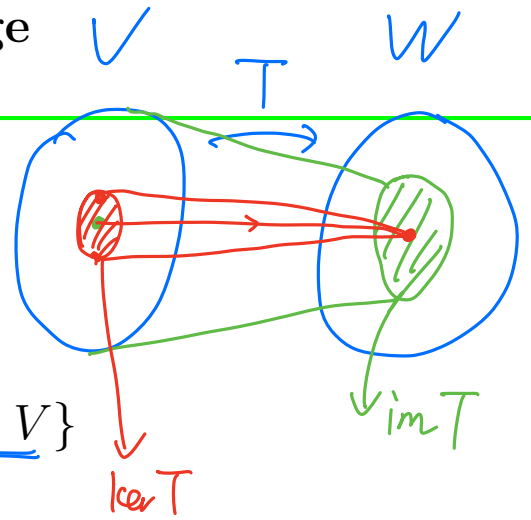
$$T: V \rightarrow W.$$

The **image** of T is defined as

$$\text{im}(T) := \{T(\vec{x}) \mid \text{all } \vec{x} \in V\}$$

The **kernel** of T is defined as

$$\text{ker}(T) := \{\vec{x} \in V \mid T(\vec{x}) = \vec{0}\}$$



Proposition 24. $T: V \rightarrow W$ is injective if and only if $\text{ker}(T) = \{\vec{0}\}$.
 $T: V \rightarrow W$ is surjective if and only if $\text{im}(T) = W$.

Theorem 25. Let $T: V \rightarrow W$ be a linear transformation. Then $\text{im}(T)$ is a subspace of W and $\text{ker}(T)$ is a subspace of V .

Check, $\vec{0} \in \text{ker}(T)$

2. closed for +

3. closed for scalar prod.

$$T(\vec{0}_V) = \vec{0}_W \checkmark \Rightarrow \vec{0}_V \in \text{ker } T$$

$$2. \text{ For any } \vec{u}, \vec{v} \in \text{ker } T, \Rightarrow T(\vec{u}) = \vec{0}_W, T(\vec{v}) = \vec{0}_W$$

$$T(\vec{u} + \vec{v}) \stackrel{\text{linear}}{=} T(\vec{u}) + T(\vec{v}) = \vec{0}_W + \vec{0}_W = \vec{0}_W \Rightarrow \vec{u} + \vec{v} \in \text{ker } T$$

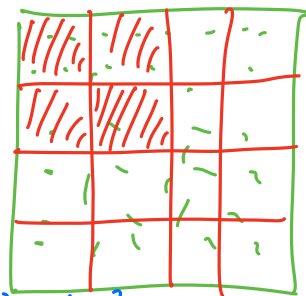
$$3. \text{ For any } c \in \mathbb{R}, \vec{u} \in \text{ker } T \Rightarrow T(c\vec{u}) = cT(\vec{u}) = c\vec{0}_W = \vec{0}_W$$

$$\mathbb{Z} = \{1, -1, 2, -2, 0, 1, 2, 3, \dots\}$$

$\mathbb{Z}_2 = \left\{ \begin{matrix} \text{even} \\ [0] \end{matrix}, \begin{matrix} \text{odd} \\ [1] \end{matrix} \right\} = \mathbb{Z}/2\mathbb{Z}$ 5. Quotient spaces.

An **equivalent relation** $\sim$ on a set V is a binary relation such that for any $\vec{u}, \vec{v}, \vec{w} \in V$,

- $\vec{v} \sim \vec{v}$. *reflexive*
- If $\vec{v} \sim \vec{w}$, then $\vec{w} \sim \vec{v}$. *symmetric*
- If $\vec{u} \sim \vec{v}$ and $\vec{v} \sim \vec{w}$, then $\vec{u} \sim \vec{w}$. *transitive*



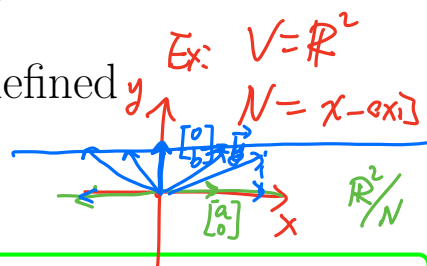
Let V be a vector space over a field $\mathbb{F}$. Let N be a subspace of V . We can define an equivalence relation on V by defining that

$$\vec{v} \sim \vec{w} \text{ if and only if } \vec{v} - \vec{w} \in N$$

The **equivalence class** (or, called the **coset**) of $\vec{v}$ is defined

$$[\vec{v}] := \vec{v} + N = \{\vec{v} + \vec{a} \mid \vec{a} \in N\}$$

$$[0] = 0 + 2\mathbb{Z} = \{0, \pm 2, \pm 4, \dots\}$$



Definition 26. The **quotient space** V/N is the set of all cosets.

Sum and scalar product are defined as

- $[\vec{v}] + [\vec{w}] = [\vec{v} + \vec{w}]$.
- $c[\vec{v}] = [c\vec{v}]$.

isomorph

$$\mathbb{R} \cong V/N = \{[0], [\vec{y}]\}$$

$[0] = N$
 $[\vec{y}] = \vec{y} + N$
 $\vec{y} = [0]$

Proposition 27. Quotient space V/N is a vector space.

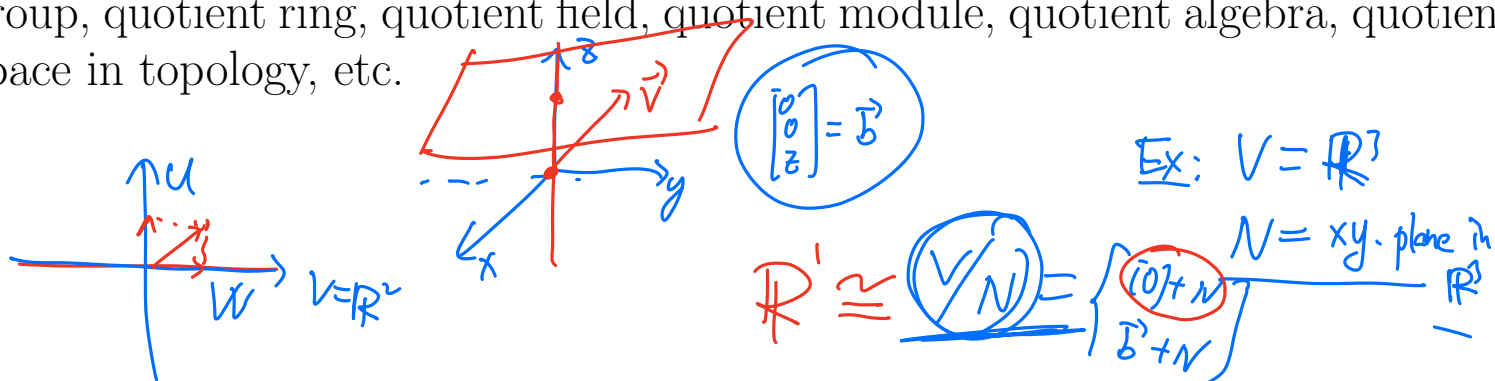
There is a natural epimorphism from $p : V \rightarrow V/N$ defined by $p(\vec{v}) = [\vec{v}]$. The kernel is $\ker p = N$. There exists a **short exact sequence**

$$0 \rightarrow N \xrightarrow{f} V \xrightarrow{p} V/N \rightarrow 0$$

ker p = im f

Here, exact means $\ker = \text{im}$ at each connecting place.

Remark: The idea of quotient is used in almost all mathematics, e.g., quotient group, quotient ring, quotient field, quotient module, quotient algebra, quotient space in topology, etc.



Q: Is $U \cup W$ a subspace of V ? NO!

6. Sum and direct sum of subspaces

Q: Is $U \cap W$

a subspace of V ?
Yes

Let U and W be subspaces of a vector space $V = \mathbb{R}^n$ or $\mathbb{R}^{n \times m}$

Definition 28. The **sum** of U and W is defined as

$$U + W = \{ \vec{u} + \vec{w} \mid \text{all } \vec{u} \in U, \vec{w} \in W \} = \text{span}\{U, W\}$$

Proposition 29. $U + W$ is a subspace of V .

Definition 30. A sum $V = U + W$ is called the **direct sum** of U and W , denoted by

$$\vec{v} = \vec{u}_1 + \vec{w}_1$$

$$V = U \oplus W$$

if each $v \in V$ can be **uniquely** written as $v = u + w$.

Example 31. $U = \{ \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} \} \subset \mathbb{R}^3$ and $V = \{ \begin{bmatrix} 0 \\ y \\ z \end{bmatrix} \} \subset \mathbb{R}^3$, then $U + V = \mathbb{R}^3$.
However, $\mathbb{R}^3$ is not a direct sum of U and V .

$$\mathbb{R}^3 \ni \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ z \end{bmatrix} \in U + V$$

$$= \begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ y \\ z \end{bmatrix} \in U + V$$

Example 32. $\mathbb{R}^3$ is a direct sum of $U = \{ \begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix} \} \subset \mathbb{R}^3$ and $V = \{ \begin{bmatrix} 0 \\ y \\ z \end{bmatrix} \} \subset \mathbb{R}^3$.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ y \\ z \end{bmatrix}$$

$$\vec{0} = \vec{0} + \vec{0}$$

Proposition 33. Let $V = U + W$. If $\vec{0} = \vec{u} + \vec{w}$ implies $\vec{u} = \vec{w} = \vec{0}$, then

$$V = U \oplus W$$

$\vec{0}$ has a unique decomposition

$$\vec{v} = \vec{u}_1 + \vec{w}_1 = \vec{u}_2 + \vec{w}_2$$

$$(\vec{u}_1 - \vec{u}_2) + (\vec{w}_1 - \vec{w}_2) = \vec{0}$$

Proposition 34. $V = U \oplus W$ if and only if $V = U + W$ and $U \cap W = \{\vec{0}\}$.

7. Linear transformations and matrices

Theorem 35. Given an $m \times n$ matrix A . There is a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined as

$$T_A(\vec{x}) = A\vec{x}$$

$$\begin{array}{ccc} \mathbb{R}^{m \times n} & \longrightarrow & \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m) \\ A & \rightsquigarrow & T_A \end{array}$$

Denote $\vec{e}_1, \vec{e}_2, \dots, \vec{e}_m$ be the column vectors of the identity matrix I_m . We call them the standard vectors in $\mathbb{R}^m$.

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad \dots \quad \vec{e}_m = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

The next theorem is very effective for finding the matrix for a given linear transformation.

$$\vec{x} \mapsto T(\vec{x}) = A\vec{x}$$

Theorem 36 (Transformation matrix). Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation.

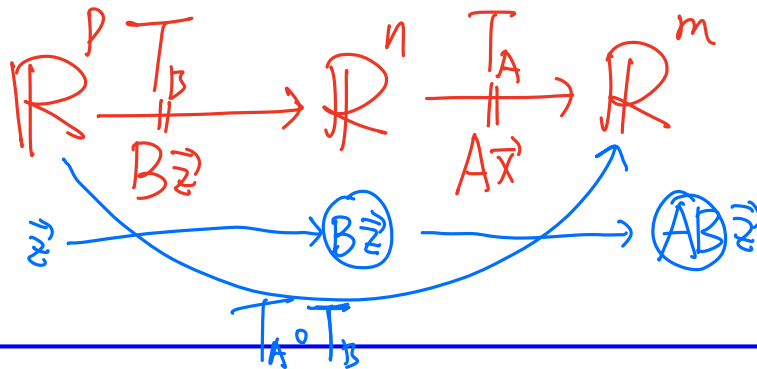
There exists an $m \times n$ matrix A such that $T(\vec{x}) = A\vec{x}$.

Furthermore, the matrix of T is given by

$$A = [T(\vec{e}_1) \quad T(\vec{e}_2) \quad \dots \quad T(\vec{e}_n)].$$

$\Leftrightarrow$ bijective

Theorem 37. An $n \times n$ matrix A is invertible if and only if the linear transformation T_A is injective; if and only if T_A is surjective.



Theorem 38. Let A be an $m \times n$ matrix and B be a $n \times p$ matrix. Then the product AB is the matrix of the transformation composition $T_A \circ T_B$.

Corollary 39. An $n \times n$ matrix A is invertible if and only if T_A is invertible. Moreover, $(T_A)^{-1} = T_{A^{-1}}$.

the matrix of $(T_A)^{-1}$ is A^{-1}